Trenitalia
ETR1000 VZI-6

Horn

Revision	Author	Date	Revision Description
00.1	R. Schiavo	27/09/2023	Draft Release Generated from DOORS module /Platform.-.Zefiro-Europe/../2F_02.07.-.Horn-VF.-.Zefiro-Europe/Horn-VF.CONCEPT, baseline 1.4 Doc 3EST000225-7905_A 2F_02.07.Zefiro-Europe.TRS.15 and 2F_02.07.Zefiro-Europe.CONCEPT.54 modified because only one switch to control the horn is present on the desk 2F_02.07.Zefiro-Europe.CONCEPT.HR01 added for horn ommand by TCMS in future implemntation
00	R. Schiavo	27/09/2023	First Emission

Content

1 Introduction 4

2 Train Configuration 4

3 Definitions and abbreviations 4

4 System Overview 5

5 System Configuration 5

6 Functional requirements 6

7 Function Horn 6

8 Function Design 6

13 Requirements 6

14 Signal Interface 8

15 I/O Interface 8

16 Supplier system 8

17 TCMS SW 8

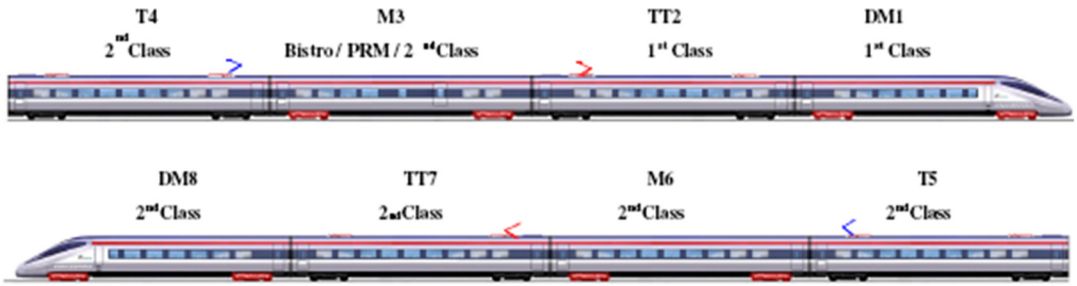
18 Bus Interface 8

1 Introduction

This document is a function specification for the control of the *Horn* system for ETR1000 / VZI-6 for customer Trenitalia.

2 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

3 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System

4 System Overview

5 System Configuration

The macrophone is used by the driver to emit a warning signal. It is controlled via 2 switches placed on the drivers desk which work only the macrophones mounted at the attached front end. There is also one pneumatic emergency button that can be operated also when electricity is not available.

6 Functional requirements

7 Function Horn

8 Function Design

Two pneumatic macrophones with two different pitches will be mounted in each front end of a train. The air supply will be controlled via magnetic valves which are operated by buttons placed on the drivers desk. The use of the macrophones shall be monitored by TCMS.

There will be one pneumatic emergency button that operates with 660 Hz. The emergency button will be possible to operate without electricity.

9 *Normal condition*

Macrophone system at the disposal of the driver to send out warning signals

10 *Failure cases*

A) One macrophone is at fault
B) Both macrophones are at fault
C) No electricity available on train

11 *Failure detection*

A) Detected during routine test carried out when the train is powered up.
B) Detected during routine test carried out when the train is powered up.
C) Detected by the driver

12 *Failure handling*

A) Train available for unlimited service. Repair during next depot visit.
B) Train out of service. Repair during corrective maintenance.
C) The driver can use the emergency button for macrophone.

13 Requirements

ID	2F_02.07.Zefiro-Europe.TRS.13
Description	The horn shall be pneumatically operated.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.07.Zefiro-Europe.CONCEPT.53	The horn shall be pneumatically operated	Derived Requirement	4S_02.01.05.-.Horns and Macrofons		N/A
2F_02.07.Zefiro-Europe.CONCEPT.69	For each horn the air supply shall be controlled by a separate magnetic valve.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_02.07.Zefiro-Europe.CONCEPT.87	The high frequency horn shall be additionaly controlled via a pure pneumatic system connected to an emergency button	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_02.07.Zefiro-Europe.CONCEPT.70	Each horn shall be separatly blockable by a cut-off cock.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A

ID	2F_02.07.Zefiro-Europe.TRS.11
Description	The horn shall be activated by a control in the driver´s desk.
Safety level	SIL 0

ID	2F_02.07.Zefiro-Europe.TRS.15
Description	There shall be a control in the cab to operate the horn
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.07.Zefiro-Europe.CONCEPT.54	A three-position-switch (one for driver one for second driver) shall be positioned on the drivers desk with the functions	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A

	– low frequency horn(unstable), – horn off (stable) – high frequency horn(unstable)				
2F_02.07.Zefiro-Europe.CONCEPT.79	A pneumatic emergency button shall be installed on the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A

The emergency push button shall be connected to the high frequency horn.

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.07.Zefiro-Europe.CONCEPT.35	The magnetic valve of the high frequency horn shall be connected to the switch on the drivers desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_02.07.Zefiro-Europe.CONCEPT.36	The magnetic valve of the low frequency horn shall be connected to the switches on the drivers desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_02.07.Zefiro-Europe.CONCEPT.71	The activation of the horns shall be monitored by the juridical recorder. (Electrical switches and emergency button)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.07.Zefiro-Europe.CONCEPT.92	Horn control elements (Electrical switches and emergency button) shall have an electrical link to the juridical recorder.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.07.Zefiro-Europe.CONCEPT.74	The horn control switches shall be active when battery power is available	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_02.07.Zefiro-Europe.TRS.14
Description	When activating the emergency stop the horn shall automatically sound.
Safety level	SIL 0

14 Signal Interface

15 I/O Interface

16 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.28	n/a		

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.29	n/a		

17 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.30	Switches and pneumatic emergency button for horn activation used.	L	2F_02.07.Zefiro-Europe.CONCEPT.71

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.HR01	TCMS_Horn_com for future implementations	L	

18 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.32	n/a		

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.33	n/a		